

Lesson 5: Applying Color Filters to Images

How to Filter a Two-Tone Image with MATLAB

Read and Show a Two-Tone Image

The fileName below links to a two-tone image. Or, with your sidekicks permission, feel free to find your own on the internet by searching for "red blue double exposure" images. Then, while wearing 3D glasses, try closing one eye at a time....

```
fileName = "https://freight.cargo.site/w/1340/i/a36a16cdf5722226d49a4a9f5797" + ...  
           "a2863215d1f1da0d1d08f4064bb6c2442582/13-046-lynx-BETES-DE-MODE-DIMETTO_o.jpg";  
I = imread(fileName);  
imshow(I)
```

We'll use MATLAB to filter out colors just like the 3D glasses. To do so, we'll be using the newFilter function from the [Pixels to Pictures](#) courseware.

Create Color Filters

In order to use these filters they must be the same size as our image. We can use the size function to determine the size of an image (or array). This function returns an array with values for the size along each dimension.

```
sz = size(I)
```

Recall, images have three dimensions: rows, columns, and pages. So the variable sz contains the number of rows, the number of columns, and the number of pages, respectively. We can then extract these values using what we previously learned about indexing.

```
rows = sz(1);  
cols = sz(2);
```

We'll set the RGB values of these filters to match the two colors of our 3D glasses: red and cyan (green/blue). Recall we specify colors using Red, Green, and Blue (RGB) values, which range from 0 to 255.

```
red = [255 0 0];  
cyan = [0 255 255];  
R = newFilter(rows,cols,red);  
C = newFilter(rows,cols,cyan);  
imshowpair(R,C,'montage')
```

Use Filters to Remove Opposite Color from Each Image

We can extract the cyan-toned image by filtering out all the red shades; and we can extract the red-tone image by filtering out all the cyan shades.

```
Icyan = I - R; % remove red  
Ired = I - C; % remove cyan
```

```
imshowpair(Icyan,Ired,'montage')
```